

Rayhaan Farooq

 rayhaanfarooq.com  705-970-7294  rayhaanfarooq@outlook.com  linkedin.com/in/rayhaanfarooq  github.com/rayhaanfarooq

EDUCATION

Carleton University - Honours BSc Computer Science

September 2023 - April 2028

Concentration: AI/ML | CGPA: 3.95/4.00 | Co-op | Fourth-Year Standing (Senior)

Ottawa, ON

WORK EXPERIENCE

Fullscript | *Elixir, Ruby on Rails, GraphQL, MySQL, OpenSearch, Langfuse, Jido*

May 2026 – December 2026

Software Engineer Intern: AI Agents & Search Engine

Toronto, ON

- Led the design and implementation of Storefront Protocols, integrating AI capabilities into a marketing tool used by **125,000** practitioners by defining product requirements, designing workflows, and partnering across engineering.
- Built an end-to-end clinical research pipeline for **Assist (AI Agent)** using **Elixir**, NatMed APIs, concurrent retrieval, and **Top-K** ranking, reducing clinical evidence tokens by **80%** and follow-up research latency by **5s**.
- Architected reusable agentic infrastructure for **Assist (AI Agent)** using synthetic prompts, scoped skills/tools, and reachability guards, supporting **2** distinct onboarding workflows with deterministic, context-aware agent behavior.
- Replaced a verbose MCP tool with an **OpenSearch**-backed search interface, reducing biomarker tool payloads by **95%** and eliminating a redundant tool from every agent prompt.
- Expanded LLM observability by instrumenting **3** untracked generation workflows with **OpenTelemetry** and **Langfuse**, enabling cost attribution, token analytics, and production tracing across **15+** telemetry attributes per request.

Shopify | *TypeScript, React.js, MySQL, Ruby, GraphQL, Bootstrap, GCP*

September 2025 – December 2025

Software Engineer Intern: Talent Engineering

Toronto, ON

- Reduced administrative workload by **10×** by building a full-stack Rails platform that replaced **15+** Excel spreadsheets with automated workflows, dashboards, and centralized data, supporting **300+** interns and **40+** Early Careers staff.
- Designed **relational data models (ERDs)** and integrated **BigQuery** to centralize **2,000+** intern records, enabling scalable analytics and data access for **75+** managers and **200+** mentors.
- Implemented a **Gale-Shapley** stable matching engine that automated **80%** of intern-to-team allocations, reducing manual assignment time from **4 weeks** to under **10 minutes**.

Ross Video | *Python, TypeScript, React.js, SQL, Docker, Jenkins, AWS, Go, Java*

May 2025 – August 2025

Software Engineer Intern: AI Tools & Test Automation

Toronto, ON

- Built an internal **RAG AI chatbot** using **React**, **TypeScript**, **Express.js**, and **PostgreSQL**, reducing deployment and environment setup time by over **80%** through containerized deployments on a custom **VM**.
- Built an automated **RAG ingestion pipeline** in **Python** and **Java** with **Jenkins**, processing **100+** documents into embeddings stored in **AWS S3**, reducing manual document processing by over **90%**.
- Collaborated with cross-functional engineering teams to deploy the chatbot across departments, improving access to internal documentation for **20+** employees and reducing onboarding time by up to **50%**.

Carleton University | *Java*

September 2024 – April 2025 & January 2026 – April 2026

Teaching Assistant: Abstract Data Structures & Algorithms

Ottawa, ON

- Led tutorials and mentorship for **200+** students in **Java**, **OOP**, and **algorithms**, boosting average exam scores by **20%** and assignment performance by up to **25%** through targeted problem design and structured feedback.

PERSONAL PROJECTS

WhatsUpMarket (Live Website) | *Python, Alpha Vantage API, Discord Webhooks, Git, Cloudflare, PostgreSQL*

September 2026

- Built a market intelligence platform serving **1,000+ users** with real-time trading data and news via **Alpha Vantage API**.
- Built **Python ML classifiers** for financial news and integrated **Cloudflare Workers** and **Discord webhooks** for alerts.

Jobber | *Go, Cloudflare Workers, Discord API*

November 2025

- Built a job alert system detecting postings and delivering **Discord** alerts, growing a **1000+** member community.
- Engineered a self-hosted backend on a **Raspberry Pi** using **Go** and **Cloudflare Workers** for alert delivery.

TECHNICAL EXPERTISE

Languages: TypeScript, JavaScript, Python, Go, Java, Ruby, C++, C, SQL, Elixir, Haskell, Rust

Frameworks/Libraries: React.js, Next.js, Node.js, Ruby on Rails, Phoenix, TailwindCSS, GraphQL

Databases/Cloud: PostgreSQL, MySQL, Redis, Kafka, AWS, GCP, Kubernetes, OpenSearch, Turbopuffer, Elasticsearch

Developer Tools: Docker, Git, Jenkins, Linux, QNX, Grafana, OpenTelemetry, Stripe, Cursor, Claude Code, Linear, Jira

LEADERSHIP / EXTRACURRICULAR

cuHacking & Hack The Hill

November 2023 – November 2025

2× Sponsorship Coordinator

Ottawa, ON

- Raised **\$110K+** from **20+** sponsors across **cuHacking** and **Hack The Hill**, leading two of Ottawa's largest hackathons with **1,400+** attendees and driving end-to-end execution and sponsor success while leading a team of **50+** people.